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AI Resilience Framework

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AI Resilience Framework

1. MEASURING RESILIENCE

Resilience is largely defined as the ability to adapt or recover from adverse conditions, stresses, attacks, or compromises on systems that use or are enabled by digital resources (https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-160v2r1.pdf). In AMARANTH, resilience is measured in the amount of time it took from the beginning of a testing period for the model to reach predictions outside of the original 95% confidence interval and the Kullback-Leibler (KL) divergence theorem. The Model Agnostic Prediction Interval Estimator (MAPIE) (https://mapie.readthedocs.io/en/stable/) modeling architecture was used to calculate confidence intervals using the Jackknife+ approach where (https://levelvc.com/jackknife-and-model-confidence/):

For a set of n points in $\mathbb{R}^k$ a model $\mu: \mathbb{R}^k \to \mathbb{R}$ is trying to forecast scalar values. For a particular datapoint $i \in \{1, \dots, n\} \subset \mathbb{N}$ the model is minimizing $\mu(x_i) - y_i$ for feature datapoint x_i and target value y_i.

At train time, Jackknife+ approach trains models μ_i on every point except for i and then computes the residual for each model on each point:

$$R_i \triangleq |\mu_i(x_i) - y_i| \tag{1}$$

At inference for a new point x_{n+1}, the model is used to predict the target value according to $\mu_i(x_{n+1})$ with the upper and lower residuals calculated via:

$$R_{\pm} \triangleq \mu_i(x_{n+1}) \pm R_i. \tag{2}$$

Given a confidence interval, value $\alpha = 1 - (confidence\ interval\ \%)$, and the lower bound is calculated as:

$$v_{-} = \alpha(n+1) \text{ and the upper bound as } v_{+} = (1-a)(n+1) \tag{3}$$

for the smallest value in R_{-} and R_{+} respectively. Therefore, the confidence intervals are (https://www.stat.cmu.edu/~ryantibs/papers/jackknife.pdf):

$$C(x_{n+1}) = \left[v_{-}\{\mu_i(x_{n+1}) - R_{i,-}\}, v_{+}\{\mu_i(x_{n+1}) + R_{i,+}\}\right] \tag{4}$$

We can use these confidence intervals to figure out when $\mu_i\ (x_{n+1})$ exceeds 95% confidence established during the testing phase, such that $C(x_{n+1})_{-} > \mu_i\ (x_{n+1})$ or $C(x_{n+1})_{+} < \mu_i(x_n + 1)$. In a regression model using timeseries data such as the datasets for this project, the "point of failure" (PF), or the time it takes for the model to drift outside of an unacceptable range established as the time it takes for the model to first drift out of the confidence interval, where at x_{n+2} moderate drift occurs minus the first time in deployment x_{n+1}. This assumption assumes no recovery following x_{n+2} as a conservative estimation.

$$PF = C(x_{n+2})_{+} - C(x_{n+2})_{-} > C(x_{n+1})_{+} - C(x_{n+1})_{-}$$

This formulation assumes no recovery after x_{n+2}as a conservative estimation. Calculating P_F through confidence intervals is beneficial when there are significant variations in data both in volume and magnitude. Calculating the confidence interval generally is dependent both on the model and the data, making it a good estimation of model drift for specific applications.
Despite this, calculating P_F through confidence intervals can be extremely conservative in drift estimation where no discernible drift can be visually identified at P_F. Additionally, the calculation of P_F through confidence intervals requires a supervised or semi-supervised approach to calculate the residuals.

(5)

Using an example of load forecasting models for Austin Texas trained on weather and load data from 2002 to 2014, testing drift from 2014 to 2016, and assuming deployment from 2016 onward, the Jackknife derived P_F can be shown to be extremely conservative in drift estimation. For example, as shown below, with a Seasonal Autoregressive Integrated Moving Average (SARIMA) model traditionally used for load forecasting gradually gets a larger confidence interval and underpredicts the load, where the grey dashed line represents the P_F —significantly under-predicting when model drift occurs, where the P_F is only 762 days from the date of deployment. At this point, no discernible drift can be visually identified. To compensate for this, a different PF method was introduced—KL divergence to calculate the distance between a true probability distribution and the model's probability distribution, where the KL PF point is denoted as a green dash-dot line in Figure 1, known as the moderate drift point of 1886 days from deployment. The red dash-dot line represents the significant drift point.

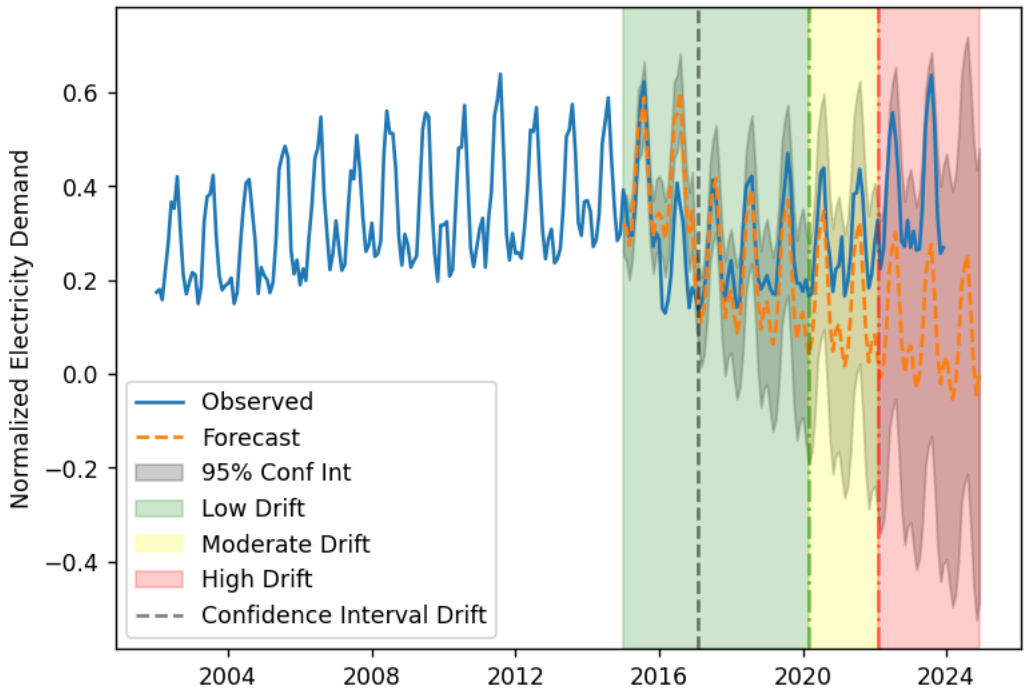

Figure 1. SARIMA forecasted normalized electricity demand.

KL divergence can be calculated as:

$$D_{KL}(P(y_i)||Q(\mu_i(x_i))) = \begin{cases} P(y_i)\log\dfrac{P(y_i)}{Q(\mu_i(x_i))} - P(y_i) + Q(\mu_i(x_i)) & \mu_i(x_i) > 0, y_i > 0 \\ y_i & \mu_i(x_i) = 0, y_i \geq 0 \\ \infty & otherwise \end{cases} \tag{6}$$

where

$Q(\mu_i(x_i))$ = model probability distribution

$P(y_i)$ = true probability distribution.

The total entropy of a system is calculated as:

$$D_{KL}(P||Q) = \sum_{i}^{\mathbb{N}} D_{KL}(P(y_i)||Q(\mu_i(x_i)))$$

For a well-fitting model, $D_{KL}\ (P\ ||\ Q\ )$ should be less than 0.25 (https://link.springer.com/article/10.1007/s42488-024-00119-y). Below shows the KL-divergence graph, where discontinuous portions are values of $D_{KL}\ (P\ ||\ Q\ ) = \infty$. Consistent with Figure 1, the KL PF point is denoted as a yellow dash-dot line in Figure 2, known as the moderate drift point ($0.1 < D_{KL}\ (P\ ||Q\ ) < 0.25$). The red dash-dot line represents the significant drift point ($D_{KL}\ (P\ ||Q\ ) > 0.25$), and the green dash-dot line represents the low drift point ($D_{KL}\ (P\ ||Q\ ) < 0.1$) (https://digitalcommons.chapman.edu/cgi/viewcontent.cgi?article=1199&context=engineering_articles).

For simplicity, we consider $P_F = x_{n+2}$ to be at the point where $D_{KL}(P(y_i)|\left|Q\left(\mu_i(x_i)\right)\right) > 0.1$ for the first time following deployment x_{n+1}.

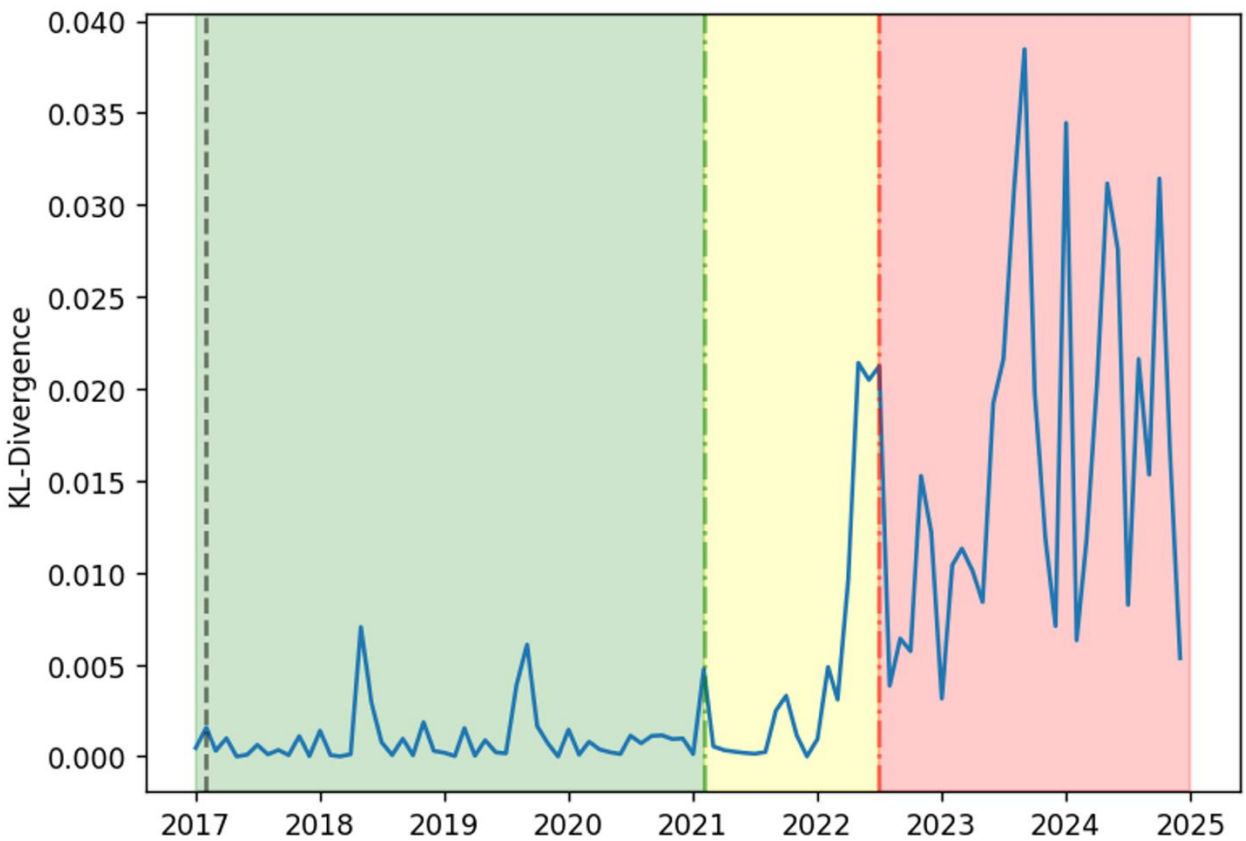

Figure 2. KL-divergence of SARIMA model 2017-2025.

Overall, the KL-divergence method works well for determining the model drift, where moderate drift can be considered the PF and when the model should be retrained, roughly mapping to a significant increase in the area of the confidence interval, as shown visually in Figure 1 and Figure 2.